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The role of competing timescales of viscoelastic stress redistribution and edge-initiated interfacial cracking in creep failure modes of pressure-sensitive adhesive joints

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ABSTRACT

It is essential to know and understand the interaction between viscoelastic behaviour and interfacial defects in the long-term failure of soft adhesive joints. Investigation of single-lap joints (SLJs) with a pressure-sensitive adhesive (PSA) acrylic revealed mechanical competition between viscoelastic stress redistribution and crack initiation due to interfacial defects. The results of creep tests on PSA joints revealed three failure modes: instantaneous interfacial failure from the edge, progressive damage from the edge, and bulk creep failure, each occurring on very different time scales. Two different time scales were identified as part of the viscoelastic behaviour of the material through stress relaxation analysis, where it was also observed that failures occurring before stress redistribution are similar to joints with poor adhesion; on the other hand, those that proved to be more resistant can withstand up to the longest relaxation time scale. By examining the edge of the SLJs before the test using scanning electron microscopy (SEM), it was possible to observe small imperfections invisible to the naked eye, located at critical points of stress concentration, which were later correlated with how the joint fails.

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creep; interfacial defects; Pressure-sensitive adhesives; stress redistribution; single-lap joints; viscoelastic failure mechanisms

1. Introduction

Pressure-sensitive adhesives (PSAs) are increasingly used in diverse industrial sectors, notably in electronics,^[1,2] where their capacity to form rapid bonds under minimal pressure enhances manufacturing efficiency. However, failure mechanisms under sustained loading remain poorly understood. Among PSA categories – acrylic,^[3] rubber,^[4] silicone,^[5] and urethane-based^[6] – acrylic PSAs are preferred for their optical clarity, photo-curing potential, and

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environmental stability.^[7] The adaptability afforded by customizable acrylic monomers explains their widespread use in applications from foldable screens to medical devices.^[2,8,9]

PSAs achieve adhesion mainly through intermolecular forces (van der Waals forces, hydrogen bonds) that form at the substrate-adhesive interface and are supplemented by their conformability, which allows them to adapt to the shape of the surface. Unlike traditional adhesives, PSAs do not rely on mechanical interlocking (its contribution is minimal); instead, their rapid bonding ability derives from their low melting point and high flexibility, which allows them to wet the surface and remain in intimate contact with the substrate without applying heat or pressure.^[10–12] Their debonding behaviour involves cavity nucleation, expansion, and fibril formation – processes that dissipate energy during separation.^[13–15] Critically, these mechanisms are time-dependent: rheological properties often dominate performance,^[16–18] and viscoelastic stress redistribution governs long-term durability under sustained load. This time-dependency makes failure prediction challenging, particularly when interfacial defects are present.

The long-term reliability of PSA joints under sustained loading presents a significant challenge, despite their widespread use. The viscoelastic behaviour of these materials, characterized by the complex interplay between their flow characteristics and elastic properties, significantly influences their time-dependent performance under load.^[10] This viscoelasticity manifests as creep behaviour – progressive deformation under constant stress – which can critically impact the service life and structural integrity of PSA joints in long-term applications.^[19–21] While creep has been studied in various soft materials,^[22–24] its mechanisms in PSA joints – particularly how interfacial quality affects failure – remain controversial.^[25] Particularly, the creep-induced failure in PSA-bonded joints is challenging to study and has received limited attention in the literature, especially for the single-lap joint (SLJ) configuration. Although simplified compared to practical bonded assemblies, SLJs are widely adopted in laboratory studies as representative models for investigating adhesive joint mechanics. They effectively capture key mechanical characteristics and stress concentrations relevant to practical bonded assemblies, rendering them a standard configuration for experimental and theoretical investigations.^[26,27] Classical stress analyses of SLJs (Volkersen,^[28] Goland-Reissner^[29]) predict stress concentrations at overlap edges in elastic materials. However, these models do not account for time-dependent viscoelastic behaviour or the influence of micro-scale interfacial defects that characterize PSA failure under sustained loading. As such, understanding failure mechanisms is of utmost importance to improve design and quality control. Studies by Geiss and Vogt^[30] employing standard tests typically used for structural adhesives revealed significant dependencies of PSA adhesion on environmental factors, particularly moisture levels. Townsend et al.^[31]

investigated high-performance acrylic PSAs as potential alternatives for structural glazing, demonstrating greater sensitivity to applied load levels compared to conventional silicone adhesives. Similarly, Ernault et al.^[32] explored creep behaviour in SLJs of hot melt PSAs, highlighting the significant influence of substrate properties on viscoelastic response and the divergent performance characteristics of semicrystalline versus amorphous adhesives under creep conditions.

Despite these advances in understanding PSA mechanics, the relationship between interfacial bonding quality and creep failure remains under-explored. Specifically, edge effects and manufacturing-induced defects at overlap edges – critical stress concentration sites – are poorly characterized in PSA joints.

Failure in soft adhesive joints often initiates at local defects or edge irregularities, where stress concentrations are highest. For PSAs specifically, this edge sensitivity is compounded by viscoelastic stress redistribution and time-dependent damage evolution. Xue et al.^[33] recently demonstrated that elastomers and PSA systems commonly fail from the edges, rather than from internal flaws, due to localized strain amplification. Similarly, Long et al.^[34] emphasized that the fracture of highly deformable soft materials is fundamentally governed by the interplay between defect nucleation and large-scale stress redistribution. In PSA systems, failure modes such as cavitation, fibrillation, and crack propagation compete depending on adhesive architecture and loading conditions, as reviewed by Sosson et al.^[35] and Bartlett et al.^[36]

In addition to inherent material defects, fabrication processes themselves may contribute to the creation of voids and localized weaknesses in PSA joints. SLJs made of PSAs are typically subjected to constant pressure during manufacturing processes to establish intimate contact between the adhesive and the substrates.^[37] Under constant pressure, gas absorption and potential saturation within the adhesive material can take place. Following the application of constant pressure, a sudden release of pressure can trigger a rapid decompression phase. If the absorbed gas cannot diffuse out quickly enough during decompression, it leads to localized expansion and the creation of cavities.^[38] This phenomenon has been documented in various polymer systems, indicating that similar processes could occur in PSAs. The combination of stress concentrations at the edges of the joint and decompression-induced void formation may create synergistic effects, accelerating the failure process. Limited studies have examined the formation of voids or voids at the edges of PSA joints. Initial observations by Yamaguchi^[39] noted the occurrence of cavitation processes during creep loading, with hydrostatic stress gradually decreasing over time due to these processes. More recent work by Chiche and Creton^[40] provided insights into the impact of surface properties on cavity formation, though primarily in the context of tack testing rather than long-term creep behaviour.

This study addresses this gap by investigating how interfacial defects – specifically their location and severity near overlap edges – govern creep failure in acrylic PSA single-lap joints. We hypothesize that edge defects control failure by preventing viscoelastic stress redistribution, causing localized damage initiation within seconds of loading. To test this, we combine creep testing ($n = 50$ joints), stress relaxation analysis, post-mortem surface quantification, and SEM microscopy with statistical modelling (OLS regression, Cox survival analysis). Our results reveal that adhesive failure percentage – a post-test marker of interfacial quality – correlates strongly with rupture time ($r = -0.67$), while conventional void metrics (pre-test optical inspection) show no predictive value. This finding challenges standard quality control approaches and demonstrates that sub-optical interfacial imperfections at stress concentration sites dominate PSA joint durability.

2. Materials and methods

2.1. Experimental design

2.1.1. Materials

The substrates used in the SLJ specimens were made of transparent polymethyl methacrylate (PMMA). This material has a Poisson's ratio of 0.37 and a Young's modulus of 3 GPa.^[41] Because of its transparency features, this material made it possible to visually inspect the damage process during testing, which allowed for a better understanding of the PSA fracture process in the SLJ tests. This enabled real-time observation of damage initiation sites and propagation patterns, critical for identifying edge-dominated failure mechanisms.

A commercial high-performance acrylic transfer tape designed for structural bonding applications was selected. This PSA exhibits temperature stability and high shear strength, making it representative of adhesives used in demanding assembly applications where long-term reliability is critical. It can be applied in industrial settings since it enables the connecting of various materials in a variety of unfavourable conditions. Regarding the properties of the PSA, it behaves almost perfectly incompressible with a Poisson's ratio of 0.49 and has a Young's modulus of 4.5×10^{-1} MPa. This align with the behaviour of high-performance acrylic PSAs and was retrieved from manufacturer reported estimations. The chemical identity of the PSA was confirmed via FTIR-ATR spectroscopy, confirming an acrylic ester-based composition consistent with the manufacturer's specifications (see Supplementary materials, Figure S1).

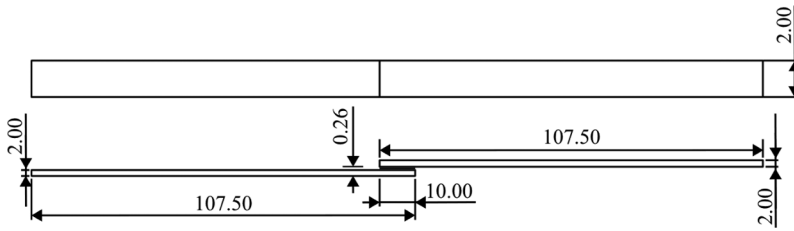


Figure 1. SLJ geometry and dimensions, in mm.

2.1.2. Specimen manufacturing

The geometry and dimensions of the SLJ joints used in this study are illustrated in **Figure 1**.

The specimen preparation started with substrate surface preparation. Substrates were cleaned with isopropyl alcohol to remove surface contaminants, and no further surface treatments were employed.

Adhesive layers, measuring 10×10 mm, were trimmed from the bulk adhesive film using scalpels to ensure dimensional precision. Subsequently, the joints were assembled within a custom-designed mould to guarantee substrate alignment and a 10 mm final overlap. Finally, a pressure of 2×10^2 kPa was applied with a weight to the overlap area for a duration of 72 h to create the best conditions for good adhesion. The applied bonding pressure (2×10^2 kPa) and subsequent rapid release upon mould removal closely replicate typical industrial lamination processes. This pressure-release cycle may contribute to gas entrapment and micro-void formation at overlap edges, as documented in polymer processing literature.^[38] The potential role of such manufacturing-induced defects in governing joint durability was a central motivation for this study.

The same specimens were adapted for further SEM analysis. For this purpose, the bonded joints were sectioned in each substrate using a precision guillotine to a length of 15 mm. Additionally, to evaluate the influence of surface preparation on interfacial contact, the edges of selected samples were ground and polished. This surface treatment was performed using a Struers Rotopol 21 grinding machine (Struers, Denmark) with 500 and 800 grit SiC abrasive papers under continuous water lubrication. The procedure aimed to improve cross-sectional visibility and assess how surface finishing affects the quality of the substrate – adhesive interface.

2.1.3. Creep testing

The creep tests were performed in an in-house built creep machine, designed to test PSAs.^[42] The setup operates on a lever-based mechanism, that allows for de-multiplication of the applied load on the specimen. Displacement was measured indirectly by retrieving voltage signals from a potentiometer, which data acquired at a rate of 10 points per second.

To define the appropriate creep load level, the maximum strength of the joints was first determined under quasi-static conditions. SLJ specimens were tested at a crosshead speed of 1 mm/min, yielding an average maximum load of (78.22 ± 6.75) N. Based on this value, the applied load for creep testing was set to 60% of the average joint strength.

A total of 50 creep tests were conducted at this load level, under ambient conditions (23°C, RT).

2.1.4. Relaxation testing

To characterize the timescales of viscoelastic stress redistribution in loaded joints, stress relaxation tests were performed on an INSTRON 8801 servo-hydraulic machine (1 kN load cell, displacement control). A rapid displacement ramp (43 mm/min to 2.3 mm final displacement) was applied to mimic the sudden loading experienced during creep testing. These parameters were derived from typical displacements observed in moderate-duration creep failures, ensuring relevance to actual loading conditions.

Five replicate tests were monitored for 120 seconds, capturing both the initial rapid stress decay and long-term plateau behaviour. A displacement of 2.3 mm (equivalent to a strain of almost $9\times$ thickness), was chosen for this research to replicate the typical deformations observed in moderate-duration creep failures (Type B, ~ 200 s). This strain threshold is above the linear viscoelastic limit but still allows the relaxation response to be close to real-world service loads. At lower strains, relaxation curves would be dominated by linear viscoelastic recovery, and would obscure the nonlinear mechanisms (e.g., initial flow, cavitation onset) that operate during failure. The defined displacement represents a deliberate compromise: sufficient to capture representative physics, but that has a short-duration (120 s) to limit irreversible failure during the test itself.

The resulting relaxation modulus was fitted with Prony series models to identify characteristic relaxation times. This analysis quantifies whether stress redistribution occurs rapidly enough (within seconds) to stabilize joints with good interfacial bonding, or whether poor edge adhesion prevents redistribution, leading to localized failure initiation.

2.1.5. SEM analysis

To evaluate interfacial quality and identify potential defects, both polished and unpolished joints were sectioned and prepared to expose the edges of the adhesive overlap region before mechanical testing. The prepared edges were then examined via SEM, using a high-resolution environmental scanning electron microscope (FEI Quanta 400 FEG ESEM/EDAX Genesis X4M) at CEMUP (University of Porto, Portugal). Images were acquired from both the corners and the middle of the overlap edges at magnifications ranging from 50x to 5000x, with an accelerating voltage of 15 kV. A total of 6 specimens

were examined: 3 reference (as-manufactured) and 3 polished joints, with 2 imaging locations per specimen (corner and mid-edge). This qualitative assessment aimed to identify the types and locations of interfacial imperfections that might escape optical detection (resolution $\sim 1.2 \mu\text{m}/\text{pixel}$) yet influence mechanical performance. This allowed for detailed documentation of surface morphology and provided qualitative insights into the bonding conditions at the adhesive-substrate interface. Potential defects of interest included voids, interfacial gaps, and areas of poor adhesive wetting.

2.1.6. Adhesive failure quantification and void area estimation

Pre-test and post-test images were acquired using a Zeiss Stemi 305 stereo-microscope equipped with a Zeiss Axiocam 208 color camera, operated via ZEN Blue 3.3 software. Images were captured at fixed optical magnification, yielding a pixel resolution of approximately $1.2 \mu\text{m}/\text{pixel}$, which allowed for consistent visualization and measurement of features such as voids, bonding quality, and overlap area.

Three interfacial characteristics were quantified for subsequent analysis:

- (1) Pre-test metrics (candidate predictors):
 - %Void area: Entrapped air voids within the adhesive overlap, visible through transparent PMMA substrates (Eq. 1).
 - Overlap area: Total bonded region.
- (2) Post-test metrics (failure outcomes):
 - %Adhesive failure: Percentage of fracture surface exhibiting clean interfacial separation versus cohesive failure (residual adhesive on both sides).

Pre-test images were analysed using ImageJ (National Institutes of Health, Bethesda, MD, USA) thresholding and area measurements. Post-test fracture surfaces were visually segmented by two independent operators, distinguishing light zones (adhesive failure) from dark zones (cohesive failure). The % adhesive failure was calculated as the area of interfacial separation divided by total bonded area.

All measurements were normalized to the overlap area to enable comparison across specimens.

Pre-test images of the transparent joints were used to identify entrapped air voids within the adhesive overlap. These appeared as circular or elliptical voids. Using thresholding and area measurements in ImageJ, the void area fraction was calculated as shown in Eq. 1:

$$\%void\ area = \left(\frac{\sum_{i=1}^n Area_{void,i}}{Area_{overlap}} \right) \times 100 \quad (1)$$

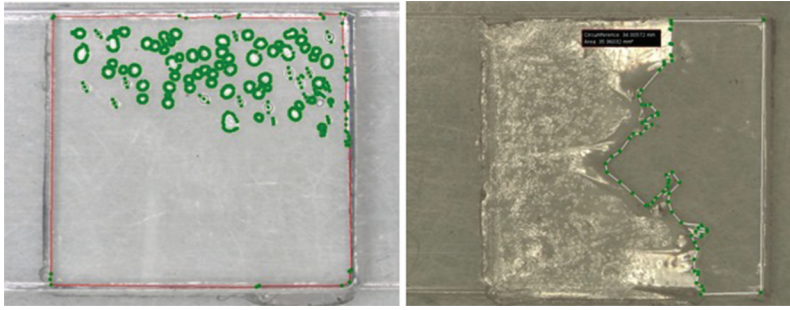


Figure 2. Digital image analysis: pre-testing %void area (left) and post-testing %adhesive failure (right).

The void area fraction was calculated as the sum of individual void areas divided by the total bonded overlap area, expressed as a percentage. The overlap area itself was also manually outlined on each specimen image. As shown in [Figure 2](#), the left panel highlights the identified voids in the overlap region, while the right panel illustrates the region used for adhesive failure estimation.

2.2. Data integration and mode classification

Rupture time measurements from creep testing ($n = 50$ joints, 46 valid) were merged with interfacial characteristics (%adhesive failure, %void area, overlap area) to create an integrated dataset for statistical analysis.

2.2.1. Failure mode classification

To identify whether distinct failure mechanisms exist, video recordings and post-test images were systematically reviewed by two independent observers without knowledge of rupture times. Three recurring damage evolution patterns were identified based on observable mechanical behaviour (as depicted in [Figure 3](#)):

Type A (Edge-initiated rapid delamination): Joints exhibiting immediate (within first 10 seconds of loading) triangular debonding from overlap corners, progressing rapidly across the interface with minimal bulk adhesive deformation. Characteristic features: sharp crack fronts, negligible fibrillation, predominantly interfacial separation.

Type B (Progressive edge-driven damage): Joints showing gradual expansion of edge voids or interfacial gaps during the first 50–100 seconds, with controlled deformation and visible stress whitening. Damage propagates progressively inward from multiple edge sites until coalescence triggers final rupture. Characteristic features: void growth sequences, mixed adhesive/cohesive zones, moderate fibrillation.

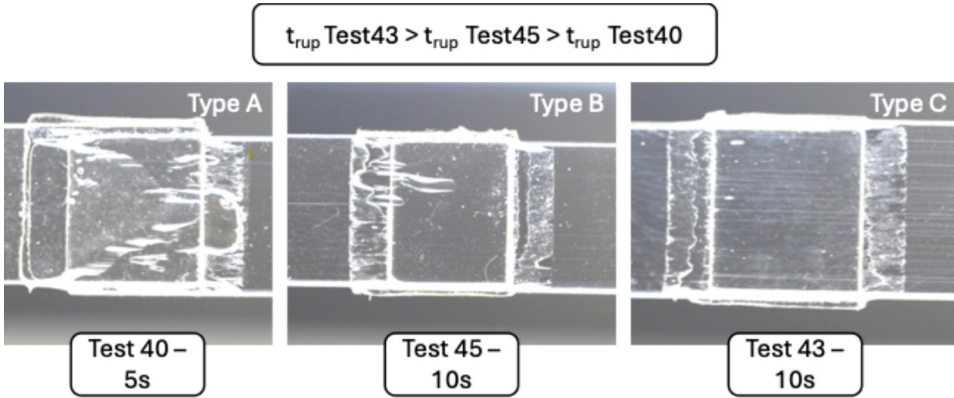


Figure 3. Representative optical images illustrating the three failure types identified through blind video analysis. Images captured at early time points (5-10s after loading) show distinct damage evolution patterns: type A (left) exhibits immediate triangular edge delamination; type B (middle) shows controlled edge void expansion; type C (right) displays uniform deformation with no visible edge damage. These mechanism-based classifications were assigned independently of rupture times, though time stamps are shown here for context. Full classification criteria are described in Section 2.2.1.

Type C (Bulk-dominated creep): Joints with no visible edge damage during initial loading phase. Failure occurs via transverse crack propagation through the adhesive bulk after prolonged creep deformation, similar to quasi-static failure mode. Characteristic features: uniform elongation, extensive fibrillation, predominantly cohesive failure surfaces.

Each test video was independently evaluated by both observers using these mechanism-based criteria. Inter-observer agreement was 91% ($\kappa = 0.87$). Borderline cases ($n = 4$) were resolved through consensus discussion prioritizing damage initiation site and propagation pattern.

2.2.2. Validation

After mechanism-based classification, rupture times for each type were examined (reported in Results). The clear temporal separation observed (Type A: median 18s; Type B: median 207s; Type C: median 581s) confirms that these mechanistically-distinct behaviours correspond to different durability outcomes, but the classification itself was established independently of time data.

All data processing and statistical analysis were conducted using Python (pandas, statsmodels, seaborn, lifelines). Specimens with incomplete or invalid data were excluded from modelling.

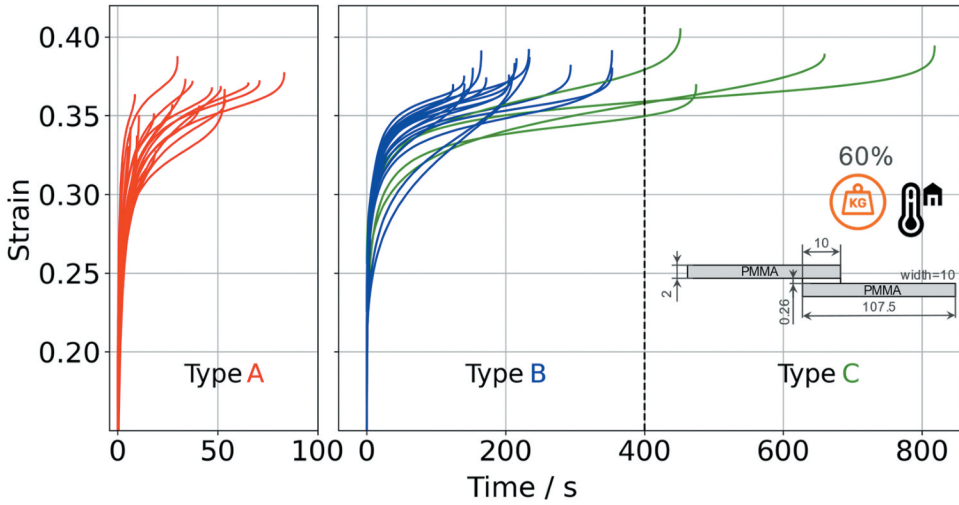


Figure 4. Experimental creep curves at 60% load level tested at RT. Curves are colour-coded by failure type: red (type A, immediate catastrophic failure), blue (type B, progressive damage from the edge), green (type C, bulk-dominated creep). The inset shows the single-lap joint geometry and loading configuration at 60% of ultimate strength. Vertical dashed lines indicate temporal boundaries between failure type regions.

3. Results

3.1. Mechanical test results

3.1.1. Creep tests

Creep tests revealed substantial variability in rupture times (range: 5–850 s, median: 132 s, [Figure 4](#)). This dispersion reflects both material viscoelasticity and the presence of interfacial defects, which influenced local strain rates and the progression of secondary and tertiary creep phases. Based on the mechanism-based classification established in Methods (Section 2.2), the 46 valid tests were distributed as follows: Type A (edge-initiated rapid delamination, $n = 23$), Type B (progressive edge-driven damage, $n = 18$), and Type C (bulk-dominated creep, $n = 5$). The temporal distribution of these mechanistically-defined types aligned with distinct rupture time ranges, validating the classification approach.

[Figure 5](#) depicts the fracture surfaces of Tests 40, 45 and 43, ordered left to right by increasing rupture time. Test 40, on the left, show evidence of catastrophic failure, by tearing and peeling the adhesive tape from the substrate surface. Also, on the bottom image, it is possible to detect some small voids in the adhesive layer or interface, consistent with cavitation, probably caused by peeling the material normal to the substrate. Test 45, on the middle, shows a different pattern, where the adhesive seems to be more deformed prior failing (clear on the bottom image), with a mix of adhesive and cohesive failure in the adhesive. However, Test 43, on the right, is the one that shows

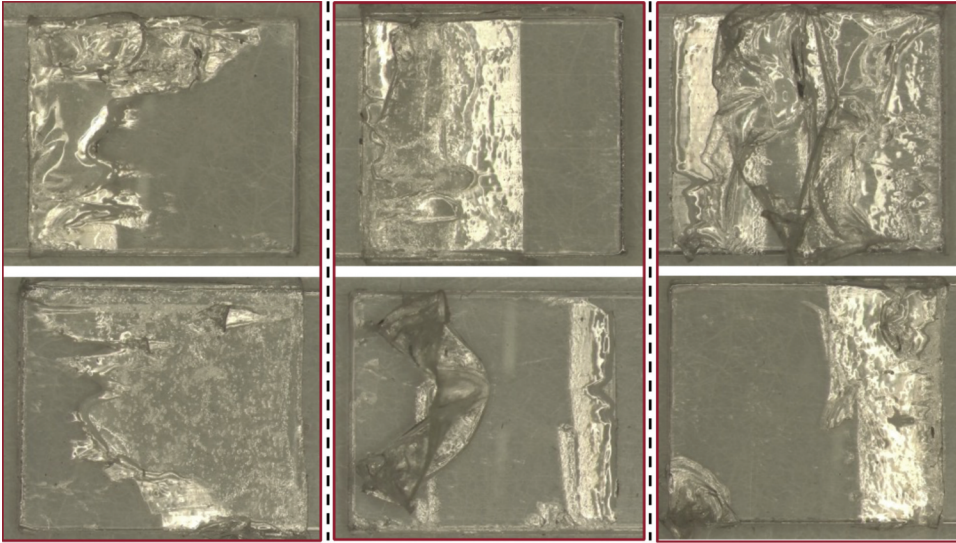


Figure 5. Fracture surfaces of test 40 (left), test 45 (middle) and test 43 (right).

more extensive material deformation, also with a mix of adhesive and cohesive failure.

The three failure types exhibited distinct temporal and mechanical signatures, consistent with their mechanism-based definitions. Type A specimens, characterized by immediate edge delamination, formed a distinctive triangular debonding pattern (visible within 10 seconds of loading) that propagated rapidly across the overlap interface, leading to ruptures within 100 seconds (median: 18s). This pattern resembles the tearing geometry investigated by reference,^[43] where the intersection of two crack fronts creates a triangular failure zone governed by film elasticity and interfacial adhesion. Type B specimens showed progressive void expansion from pre-existing edge imperfections, with controlled damage propagation leading to ruptures between 100 - 400 seconds (median: 207s). In contrast, Type C specimens

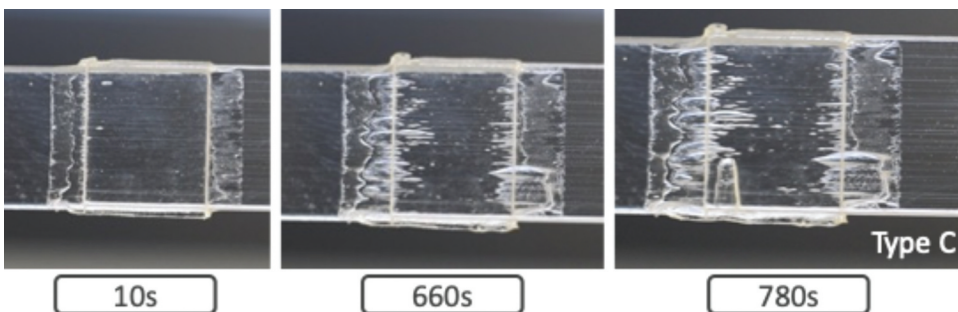


Figure 6. Test 43 damage propagation for three timestamps: after loading (left), at 80% of the rupture time (middle) and prior joint failure (right).

exhibited no visible edge damage during initial loading, with failure occurring via transverse crack propagation after prolonged deformation (>400 s), similar to quasi-static failure modes (median: 581s). [Figure 6](#) illustrates this progression for a representative Type C test, showing minimal edge damage until approximately 80% of the rupture time (660s), followed by rapid crack propagation.

3.1.2. Relaxation tests

To investigate the time-dependent mechanical behaviour of the adhesive – joint system, stress relaxation tests were performed using a displacement-controlled ramp (2.3 mm at 46 mm/min), designed to mimic loading conditions observed in Type B failures. These parameters were derived from real rupture data, ensuring that the imposed deformation generated a representative viscoelastic response. These tests aimed to quantify stress redistribution timescales following sudden loading, providing context for understanding why joints with good interfacial quality can stabilize stresses while those with edge defects fail rapidly. [Figure 7](#) illustrates the extraction of these parameters and presents representative experimental relaxation curves.

The resulting relaxation modulus was normalized and fitted using both 2-term and

3-term Prony series models to capture the decay in stress over time. The general form of the normalized Prony series used is presented in Eq. 2:

$$\frac{G(t)}{G_0} = \sum_{i=1}^N A_i e^{-t/\tau_i} \quad (2)$$

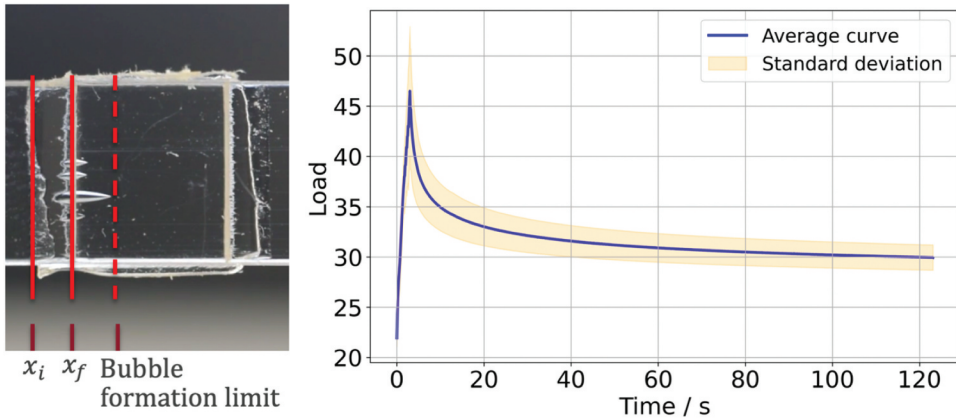


Figure 7. Extraction of experimental parameters for relaxation tests (left) and relaxation average curve in blue with standard deviation represented in yellow (right).

Where $G(t)/G_0$ is the relaxation modulus normalized to its initial value, A_i are the amplitude coefficients ($\sum A_i \approx 1$), and τ_i are the characteristic relaxation times.

The 2-term model yielded a short-term relaxation time of approximately 5.35 s and a long-term component around 1250 s, providing a strong overall match to the experimental curve. However, it struggled to fully replicate the initial steep drop in modulus, revealing the limitations of a simplified two-mode representation.

To improve fidelity, two 3-term Prony series variants were explored. The first was fitted under relaxed bounds and yielded the best visual match across the full 120-second range, especially in the early and mid-relaxation phases. However, this model required an amplitude parameter to hit the upper bound ($A_1 = 70$ MPa), suggesting potential overfitting of the short-term behaviour. To address this, a second 3-term model was constrained by more conservative, physically meaningful bounds – including a cap on amplitude ($A < 70$ MPa) and a minimum threshold for the long-term relaxation time ($\tau_3 > 1000$ s). Under these conditions, the optimizer assigned near-zero weight to the mid-range term, effectively reducing the model to two dominant relaxation components. This convergence validates the sufficiency of the 2-term model in capturing the material's behaviour while maintaining interpretability.

This analysis provides valuable insight into how stresses are redistributed after a sudden displacement is applied, which is critical for understanding the influence of local stress concentrators on joint durability, both in the short and long term. Figure 8 compares all models against the experimental curve, illustrating how increasing the number of relaxation modes can enhance fit

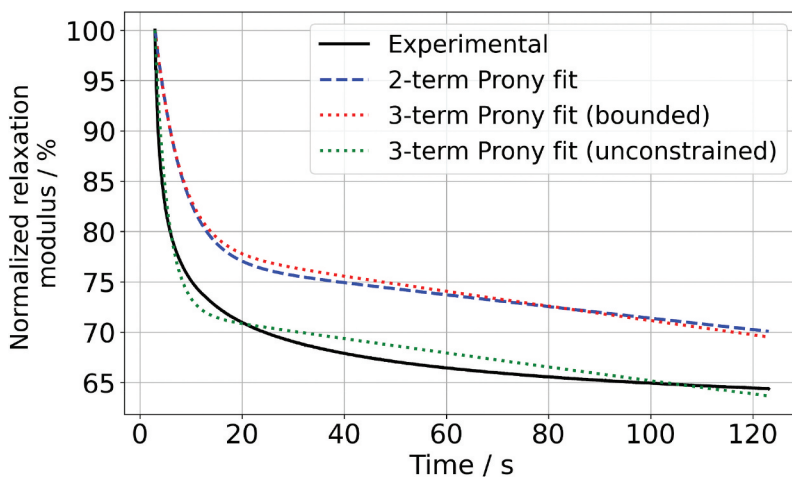


Figure 8. Normalized relaxation modulus in percentage as a function of time for experimental results and the 2-term and 3-term Prony series fit models.

accuracy – but also how model simplicity and physical realism must be balanced.

It was necessary to balance accuracy and physical interpretability when comparing two- and three-term Prony series models. When the unrestricted three-term model was used ($A_1 = 70$ MPa produced the best visual fit), the results suggested that the short-term behaviour of the model might be over-fitted. However, when the model was re-fitted with some meaningful physical constraints ($A < 70$ MPa and $\tau_3 > 1000$ s), the optimiser found that it was possible to achieve a near-zero weight for the intermediate term. These results allowed us to conclude that, in essence, the model can be reduced to a two-term model and is sufficient to capture the behaviour of the material over the 120-second observation period, which is a relevant range for joint failure time scales of 5–850 seconds.

3.2. Statistical analysis

To identify relationships between interfacial characteristics and creep durability, Pearson correlation analysis was performed on rupture time, % adhesive failure (post-test), % void area (pre-test), and overlap area. **Figure 9** presents the correlation matrix (left) and corresponding pairwise scatter plots (right).

The analysis revealed a striking disparity between pre-test and post-test metrics. The % adhesive failure, measured from fracture surfaces after testing, showed strong negative correlation with rupture time ($r = -0.67$, $p < 0.001$), indicating that specimens exhibiting predominantly interfacial separation failed earlier. In contrast, pre-test metrics showed negligible associations: % void area ($r = -0.02$, $p = 0.89$) and overlap area ($r = -0.16$,

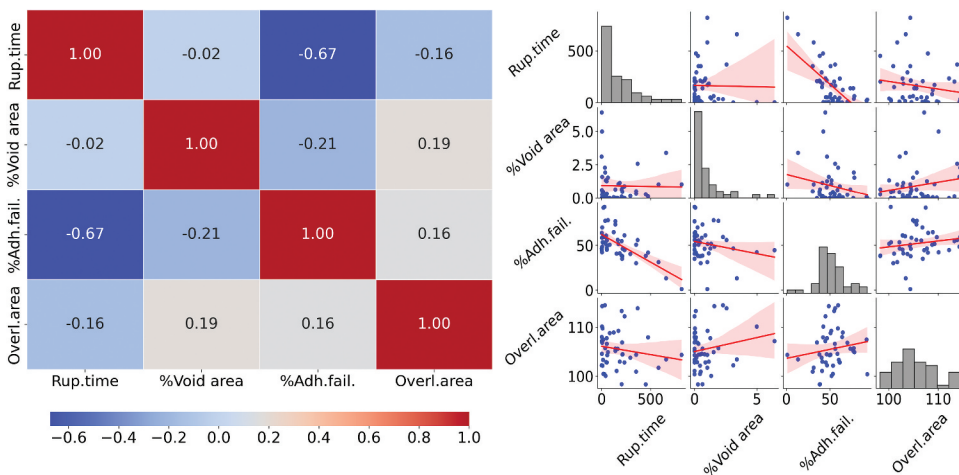


Figure 9. Pearson correlation matrix heatmap with the defined features (left) and the respective pairwise scatter plots with regression lines and confidence intervals (right).

$p = 0.28$) were essentially uncorrelated with durability. The pairwise scatter plots (Figure 9, right) illustrate this divergence, with rupture time and percentage of adhesive failure displaying a well-defined negative trend, while relationships involving pre-test void content and overlap area appear scattered and non-predictive. This finding suggests that critical interfacial defects governing failure are not detectable by standard pre-test optical inspection methods (resolution $\sim 1.2 \mu\text{m}/\text{pixel}$).

To explore whether failure surface characteristics combined with geometric features could explain rupture time variation, an Ordinary Least Squares (OLS) multiple linear regression model was constructed according to Equation 3:

$$\bar{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K \quad (3)$$

where $\bar{Y}$ is the predicted rupture time, x_i denote the explanatory variables, and β_i are the fitted coefficients. The model included % adhesive failure (post-test variable), % void area and overlap area (pre-test geometric features), and failure type (Type A/B/C, mechanism-based classification from Section 2.2) as independent variables. It has achieved strong explanatory power ($R^2 = 0.90$, adjusted $R^2 = 0.89$), indicating that these variables collectively account for most rupture time variance. Among continuous variables, only %adhesive failure reached statistical significance ($\beta = -2.02$, $p = 0.009$), with each 1% increase in interfacial failure area associated with approximately 2 seconds shorter rupture time. Pre-test void content and overlap area remained non-significant

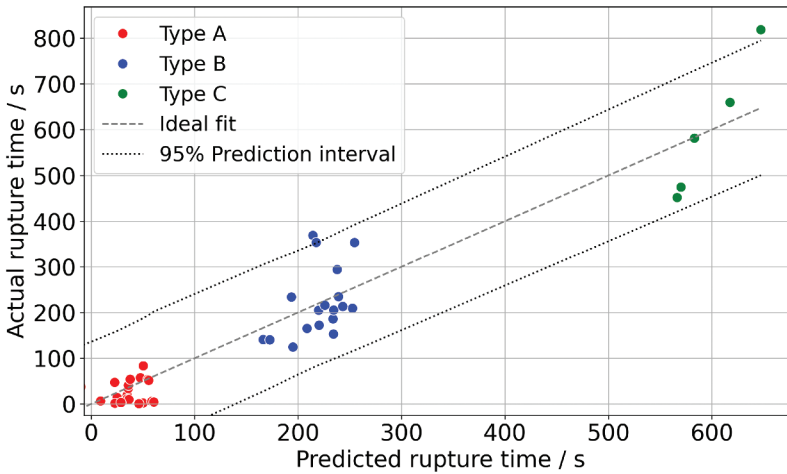


Figure 10. Comparison between predicted and actual rupture times obtained from the ordinary least squares (OLS) regression model. The dashed line represents the ideal 1:1 correspondence between predicted and observed values, while the dotted lines show the 95% prediction interval based on a LOWESS-smoothed fit.

($p > 0.6$), consistent with the correlation analysis. Failure type emerged as a highly significant categorical variable ($p < 0.001$ for Type A vs. Type C; $p < 0.001$ for Type B vs. Type C), confirming that mechanistically-distinct behaviours – defined by observable damage patterns independent of rupture time – correspond to different durability outcomes. Figure 10 shows the comparison between predicted and actual rupture times, with clear separation by failure type. It should be noted that this model is retrospective and includes post-test variables, thus quantifying associations between failure characteristics and lifetime rather than enabling prediction from pre-test inspection alone.

To account for the time-dependent nature of rupture and potential non-normality in failure time distributions, a Cox proportional hazards model was employed. This approach explicitly models rupture as a time-to-event outcome, with the hazard function defined according to Equation 4:

$$h(t|X) = h(t) \cdot \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n) \quad (4)$$

where $h(t|X)$ represents the instantaneous failure risk at time t given covariates X , $h(t)$ is the baseline hazard, and hazard ratios $\exp(\beta_i)$ indicate how each variable multiplicatively affects failure risk. Of the 50 total tests conducted, 46 valid rupture events were analysed (4 specimens excluded due to incomplete data; no censoring occurred). The model achieved good discriminatory ability with a concordance index of 0.74, meaning predicted risk ordering aligned well with observed outcomes. Consistent with OLS findings, percentage of adhesive failure showed strong association with rupture risk (HR = 1.05, $p < 0.005$), with each 1% increase in interfacial failure area corresponding to approximately 5% higher instantaneous risk of joint rupture. Pre-test metrics (percentage of void area and overlap area) again showed no significant

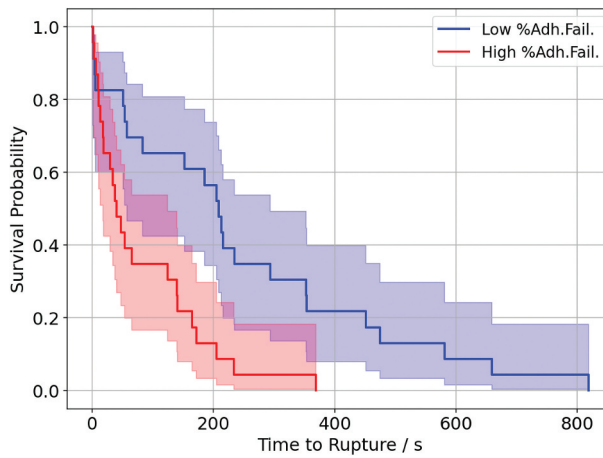


Figure 11. Kaplan–Meier survival curves for joints grouped by %adhesive failure. Specimens with adhesive failure above the median (“high %adh.Fail.”) exhibited significantly shorter survival durations under sustained load, supporting the Cox and OLS models.

association with rupture hazard ($p = 0.26$ and $p = 0.64$, respectively). The log-likelihood ratio test confirmed that the full model significantly outperforms a null (intercept-only) model ($p < 0.001$).

To visualize how interfacial quality affects rupture probability over time, Kaplan – Meier survival curves were generated. These non-parametric curves estimate the probability of a joint remaining intact up to any given time point based on observed failure events. Figure 11 presents survival curves stratified by %adhesive failure, with specimens divided into high and low groups using the median value (45%) as threshold. The curves separated clearly, with joints exhibiting high adhesive failure percentages ($>45\%$) demonstrating markedly shorter survival durations (median: 98 s) compared to low adhesive failure joints (median: 312 s). This visual result reinforces findings from both regression models.

When stratified by failure type (Figure 12), survival probabilities diverged even more dramatically. Type A (edge-initiated rapid delamination) showed the steepest decline, with median rupture time of 18 s. Type B (progressive edge-driven damage) exhibited intermediate survival (median: 207 s), while Type C (bulk-dominated creep) demonstrated substantially greater durability (median: 581 s). These patterns validate the mechanism-based classification system established in Methods Section 2.2.

To further characterize failure mode reliability, Weibull distributions were fitted to each group according to Equations 5 and 6:

$$S(t) = \exp \left[-(t/\eta)^\beta \right] \quad (5)$$

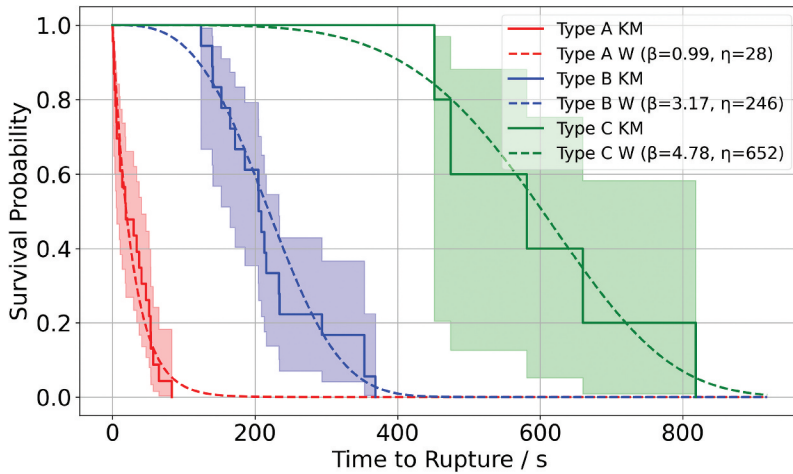


Figure 12. Survival analysis by failure mode (type A–C). Solid lines indicate Kaplan–Meier survival estimates; dashed lines represent Weibull survival models fitted to each group. Type A shows rapid, unpredictable failure; type C shows the most gradual, predictable degradation, reflecting increased reliability with decreasing interfacial damage.

$$h(t) = (\beta/\eta) \cdot (t/\eta)^{(\beta-1)} \quad (6)$$

where $S(t)$ represents survival probability, β is the shape parameter indicating rate behaviour, and η is the scale parameter representing characteristic life – the time at which approximately 63% of specimens have failed. The Weibull parameters revealed distinct reliability characteristics across failure types. Type A exhibited $\beta \approx 0.99$ and $\eta = 28$ s, indicating a near-random failure pattern ($\beta \approx 1$) driven by immediate interfacial defect activation. Type B showed $\beta \approx 3.17$ and $\eta = 246$ s, representing an increasing failure rate ($\beta > 1$) consistent with progressive damage accumulation. Type C demonstrated $\beta \approx 4.78$ and $\eta = 652$ s, indicating a strongly increasing failure rate characteristic of time-dependent degradation with predictable behaviour. The progressive increase in both scale parameters ($28 \rightarrow 246 \rightarrow 652$ s) and shape parameters ($0.99 \rightarrow 3.17 \rightarrow 4.78$) confirms that failure types represent mechanistically distinct behaviours with increasingly predictable, time-dependent characteristics. The dashed lines in [Figure 12](#) demonstrate excellent agreement between Weibull models and empirical Kaplan-Meier estimates, supporting the validity of both the classification system and the parametric characterization.

3.3. Microstructural analysis

After the previous results indicated the presence of voids, lack of adhesion, or air entrapment along the edges of the overlap, SEM analysis was conducted to further investigate these interfacial features. Since the substrates were only cleaned prior to bonding, the remaining surface roughness from the acrylic manufacturing process was hypothesized to contribute to defect formation. To explore this, selected specimens were polished along the edges of the overlap region for qualitative comparison. [Figure 13](#) presents SEM images of the overlap corner for both conditions – reference (unpolished) and polished – captured at two different magnifications.

In the reference samples ([Figure 13](#), top), clear material irregularities are visible at the corners, which could act as stress concentrators that would cause damage propagation. In contrast, the polished samples ([Figure 13](#), bottom) show a much smoother interface, suggesting improved contact between the adhesive and substrate. While polishing appears to promote better interfacial conformity, it does not inherently guarantee perfect adhesion. These observations are consistent with the initial hypothesis that decompression effects or air entrapment during the manufacturing process could contribute to defect formation at the adhesive – substrate interface, particularly near the edges of the overlap. The SEM analysis confirms that localized micro-defects can be present even in macroscopically well-bonded joints, and that surface preparation plays a role in mitigating, but not entirely eliminating, such imperfections.

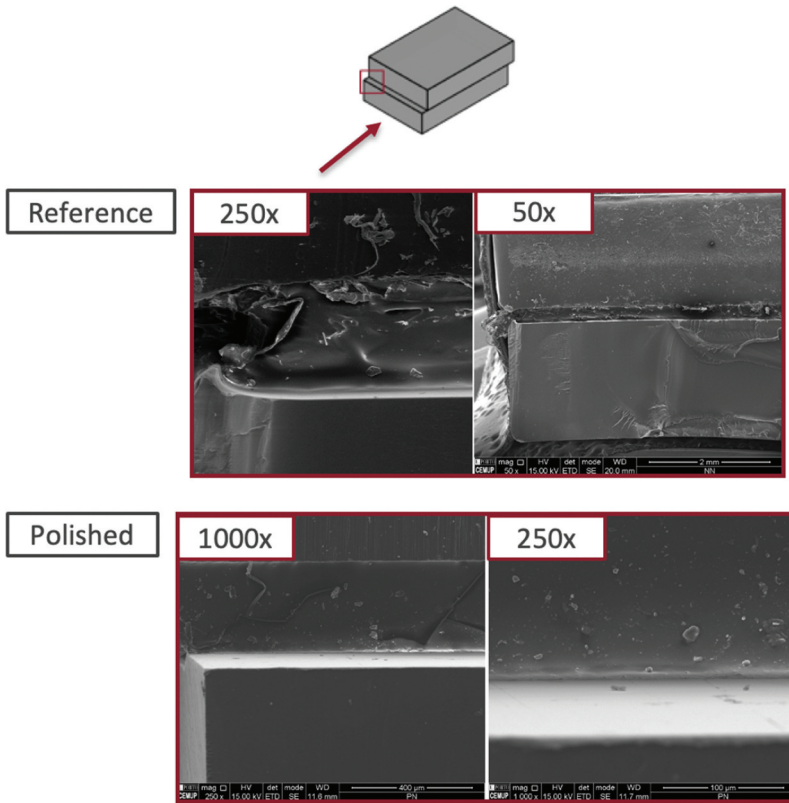


Figure 13. Cross-sectional SEM images of the adhesive joint interface before and after specimen polishing. The top row (reference) shows the unpolished joint surface at 250 $\times$ and 50 $\times$ magnifications and the bottom row (polished) at higher magnifications 1000 $\times$ and 250 $\times$. The inset schematic indicates the imaging plane (corner) relative to the bonded joint geometry.

Given that damage was observed to initiate from multiple sites during mechanical testing, a second analysis was performed at the midpoint of the overlap edge, as shown in Figure 14. The reference sample is displayed at three magnifications, and the polished sample at two. In this region, the reference condition appeared to achieve relatively uniform contact, whereas the polished condition unexpectedly exhibited discontinuities at the adhesive-substrate interface. Notably, at 1000 $\times$ magnification, some imperfections in the polished sample extended further into the overlap, potentially contributing to premature damage during creep testing.

These findings confirm that, despite apparently good macroscopic bonding, localized micro-defects may exist – particularly at the corners of the overlap – and that surface preparation (polishing) can significantly modify the interfacial quality. However, whether these defects consistently lead to early failure remains uncertain and is addressed further in the discussion.

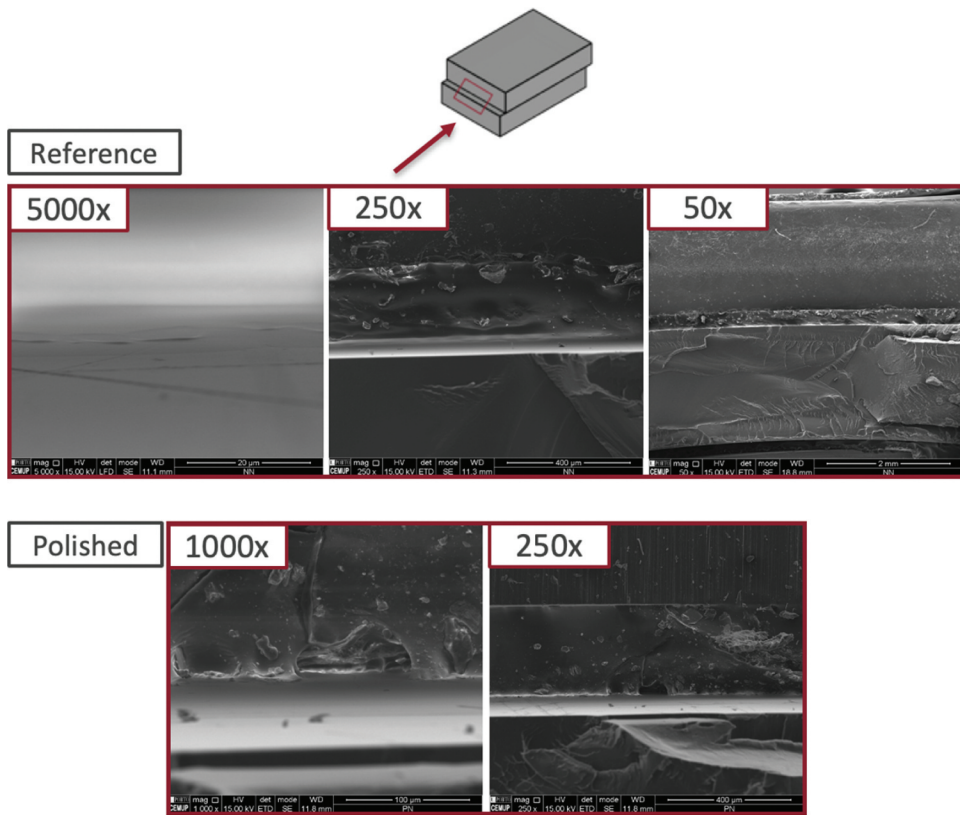


Figure 14. Cross-sectional SEM images of the adhesive joint interface before and after specimen polishing. The top row (reference) shows the unpolished joint surface at 5000 $\times$, 250 $\times$ and 50 $\times$ magnifications and the bottom row (polished) at 1000 $\times$ and 250 $\times$ magnifications. The inset schematic indicates the imaging plane (mid-plane) relative to the bonded joint geometry.

4. Discussion

Despite the widespread use of PSAs in load-bearing applications, the precise relationship between interfacial bonding quality and long-term rupture behaviour under sustained loading remains poorly understood. This study addressed this gap through a multi-scale investigation combining mechanical testing, statistical analysis, stress relaxation modelling, and microstructural observation. The findings reveal a fundamental competition between viscoelastic stress redistribution and interfacial defect-driven crack initiation that governs time-dependent failure in PSA single-lap joints. Critically, the defects controlling durability are sub-optical interfacial imperfections localized at stress concentration sites (overlap edges), not bulk-distributed voids. This mechanism explains why conventional geometric metrics such as void content ($r = -0.02$, $p = 0.89$) and overlap area ($r = -0.16$, $p = 0.28$) show no correlation with rupture time, while post-test interfacial failure characteristics –

which reflect the operative failure mechanism – strongly correlate with lifetime ($r = -0.67$, $p < 0.001$).

Video-based damage analysis, performed independently of rupture time measurements, revealed three distinct failure mechanisms arising from this competition. Type A joints exhibited immediate triangular edge delamination within the first 10 seconds of loading, progressing rapidly to failure (median: 18 s). While the short-term relaxation phase completes within ~ 5 s, Type A failures require additional time for defect-initiated crack propagation to become macroscopic (~ 18 s median), but this progression still occurs before the joint can stabilize through long-term viscoelastic mechanisms. Type B joints showed progressive void expansion from pre-existing edge imperfections, with controlled damage propagation leading to intermediate lifetimes (median: 207 s). Type C joints displayed no visible edge damage during initial loading, with failure occurring via transverse crack propagation through the adhesive bulk after prolonged deformation (median: 581 s). The clear temporal separation between these mechanistically-defined types, validated by Kaplan-Meier survival curves and Weibull analysis (Figure 12), confirms that observable damage patterns correspond to distinct durability outcomes. Critically, both OLS ($R^2 = 0.90$) and Cox regression models (concordance index = 0.74) showed that failure type significantly influenced rupture time ($p < 0.001$), demonstrating that mechanism-based classification captures real physical differences in joint behaviour rather than arbitrary temporal groupings.

The role of viscoelastic stress redistribution in determining these failure modes becomes evident through stress relaxation analysis. The two-term Prony series fit

($\tau_1 = 5.3$ s, $\tau_2 = 1250$ s) reveals two critical timescales governing mechanical response. The short-term relaxation (~ 5 s) enables rapid load redistribution in joints with good interfacial bonding, allowing uniform stress distribution across the overlap and preventing localized stress concentrations from initiating damage. However, Type A failures occurred within this timeframe (median: 18 s, range: typically < 100 s), indicating that interfacial defects at overlap edges prevent stress redistribution before stabilization occurs. The stress concentration at poorly bonded edge regions – amplified by the classical peel and shear stress distributions characteristic of single-lap joints – triggers immediate delamination when the adhesive cannot redistribute stress away from these sites. In contrast, Type C joints survive well beyond τ_2 , indicating that uniform interfacial contact enables long-term viscoelastic stabilization, with cohesive properties ultimately governing failure through bulk creep mechanisms. These findings align with Novikov et al.^[44] who demonstrated that PSA relaxation arises from multiscale processes, where short-term decay facilitates rapid stress redistribution and longer relaxation times determine the load transfer behaviour critical to cohesive joint performance. Furthermore,

Feldstein and Siegel^[45] emphasized that high-performance PSA behaviour stems from a delicate balance between cohesive strength and sufficient molecular mobility, allowing for effective stress dissipation during load application. The present results extend these concepts by demonstrating that interfacial defects – particularly at stress concentration sites – can disrupt this energy dissipation mechanism by preventing the initial stress redistribution phase. Villey et al.^[46,47] further showed that large-strain hysteresis and rate-dependent relaxation are critical for dissipating energy in soft adhesive systems, consistent with the observed competition between stress redistribution and interfacial crack propagation. Arrowood et al.^[21] similarly highlighted the role of crosslink density and bonding quality in determining defect tolerance, supporting the present observations that manufacturing-induced edge imperfections critically influence rupture behaviour. Additionally, Schrader et al.^[48] demonstrated that sustained loading conditions in soft adhesives can lead to mode I creep fracture, a behaviour consistent with the time-dependent damage accumulation observed in Type B and Type C specimens.

The statistical analysis provides quantitative evidence for this mechanism by revealing which defect characteristics govern failure. Bulk void content showed no correlation with rupture time ($r = -0.02$, $p = 0.89$), indicating that defects distributed throughout the adhesive volume do not control durability. Similarly, overlap area – a purely geometric parameter – exhibited negligible predictive value ($r = -0.16$, $p = 0.28$). In contrast, the percentage of adhesive failure measured from post-test fracture surfaces showed strong negative correlation with rupture time ($r = -0.67$, $p < 0.001$). This correlation is not circular reasoning, as adhesive failure percentage is a post-test marker reflecting which failure mechanism operated: joints failing via edge-initiated interfacial delamination (Type A) exhibit high adhesive failure percentages and short lifetimes, while those failing via bulk cohesive creep (Type C) exhibit low adhesive failure percentages and long lifetimes. The statistical relationship thus validates the proposed mechanism by demonstrating that interfacial quality at critical sites – not total defect content – determines failure mode and lifetime. This finding challenges the implicit assumption in many engineering approaches that total void content is a relevant quality metric for predicting joint durability.

The statistical analysis performed revealed a strong negative correlation ($r = -0.67$) that requires a mechanistic interpretation. The %adhesive failure, that is defined as the fraction of post-test fracture surface that exhibits a clean substrate (without adhesive residues), is not an arbitrary metric, but rather a direct indicator of failure mechanism that is operating: high %adhesive failure ($> 60\%$), short to intermediate lifetime (failure mode type A e B) or low %adhesive failure ($< 30\%$), long lifetime (failure mode type C).

In the first failure mode, joints present failure through damage with edge initiation (catastrophic failure as in type A; progressive damage from the edges

as in type B) have surfaces with predominant interfacial separation. Pre-existing defects on the edges (voids, lack of wetting, surface roughness) compromise the bonding of the adhesive to the substrate, especially in stress concentration areas. When these joints are loaded, stress concentrate on those areas that are inherently weak and lead to interfacial crack propagation; this propagation will be fast in the case of type A failure and progressive in the case of type B failure. The crack will follow pre-established weak interfaces, instead of propagating through the intact adhesive, leading to the interfacial adhesive failure of the joint.

For the second failure mode, joints with good interfacial quality will fail cohesively after prolonged deformation by creep. The failure occurs within the bulk adhesive (rupture/fibrillation of the polymer chain) instead of the interface, leading to mixed fracture surfaces (adhesive/cohesive), where there is a presence of adhesive on both substrates. The joint resists long enough for viscoelastic stabilization, enabling bulk deformation mechanism to determine the failure of the joint.

Thus, the %adhesive failure does not independently control the lifetime of the joints; instead, it is a consequence of the operational failure mode. The obtained correlation through the statistical analysis reflects a causal relationship: the interface quality determines the failure mechanisms, which in turn determines both the lifetime and the morphology of the fracture surface.

Direct microstructural evidence supports the hypothesis that sub-optical defects at stress concentration sites govern failure. SEM imaging revealed localized interfacial irregularities and micro-voids at overlap edges, particularly in unpolished specimens (Figure 13). Critically, even polished samples – which appeared defect-free under optical microscopy (resolution $\sim 1.2 \mu\text{m}/\text{pixel}$) – exhibited subtle regions of imperfect adhesive-substrate contact at the micron scale. This explains the statistical observations: optically visible voids distributed throughout the overlap do not predict failure because the critical defects are sub-optical interfacial imperfections invisible to standard inspection methods. These micro-defects, localized at overlap corners and edges where peel and shear stress concentrations are highest, appear to govern failure initiation in Type A specimens by preventing local stress redistribution. Polishing improved interfacial quality at corners and reduced visible defects, though some imperfections persisted at mid-edge locations (Figure 14), suggesting that achieving consistently defect-free joints requires not only surface preparation but also optimized bonding protocols to minimize trapped air and ensure uniform adhesive spreading. The observation that Type C joints occasionally exhibited minor interfacial defects (visible in SEM) without early failure suggests a certain defect tolerance exists when imperfections are located away from critical stress concentration zones or are sufficiently small that stress can redistribute around them within the characteristic relaxation time. This aligns with Teisseire et al.^[49] who described a competition between

cavitation and crack propagation as dominant PSA failure modes depending on defect structure, interfacial energy, and elastic modulus. In this context, Type A behaviour likely results from cavitation-type events originating at edge voids or micro-voids that amplify local stress concentrations, while Type C joints may be governed by crack propagation resisted by more robust cohesive networks after successful stress redistribution.

Synthesizing these mechanical, statistical, and microstructural observations, a comprehensive failure mechanism emerges that explains all observed phenomena. In Type A joints, sub-optical interfacial defects at overlap edges act as stress concentrators under sudden loading. The rapid stress redistribution phase ($\tau_1 \sim 5$ s) cannot stabilize the joint before localized interfacial failure initiates from these edge defects, leading to rapid triangular delamination propagation. The inability to redistribute stress away from edge concentrations overwhelms any contribution from bulk viscoelastic properties. In Type B joints, smaller, fewer, or less critically located edge defects allow partial stress redistribution. Some load transfers away from defect sites, but incomplete stress homogenization results in progressive damage accumulation over 100 – 400 s as local stress concentrations gradually drive interfacial crack growth from multiple sites. In Type C joints, good interfacial quality at edges permits complete stress redistribution during the initial loading phase. Stresses homogenize across the overlap within τ_1 , and the joint enters a stable configuration where long-term viscoelastic stabilization (governed by τ_2) allows prolonged load bearing (> 400 s). Failure ultimately occurs via cohesive viscoelastic creep mechanisms – bulk polymer chain disentanglement and flow – similar to quasi-static failure modes. This mechanism explains why percentage of adhesive failure – a post-test signature reflecting whether failure was interfacial (edge-defect dominated) or cohesive (bulk-process dominated) – strongly correlates with rupture time, while pre-test void content does not. The correlation is mechanistic, not statistical: failure mode determines both the lifetime and the fracture surface appearance.

These findings have implications beyond fundamental understanding. The dominant role of sub-optical edge defects over bulk void content suggests that characterization methods requiring higher spatial resolution (e.g., high-resolution microscopy, ultrasonic inspection at edges) or mechanical pre-screening (brief initial load observation to detect Type A behaviour) may be more informative than conventional optical void inspection for assessing joint durability. The clear temporal separation between failure modes ([Figure 12](#)) indicates that even brief initial loading observations could distinguish joints prone to rapid interfacial failure from those capable of long-term load bearing. Furthermore, the identified stress redistribution timescales provide a mechanistic basis for designing accelerated testing protocols that probe the competition between relaxation and damage initiation relevant to service conditions.

5. Conclusions

This study investigated the failure mechanisms of acrylic PSA single-lap joints under sustained loading, revealing that conventional quality control approaches based on void inspection are insufficient for predicting joint durability:

- Pre-test void content showed no correlation with rupture time ($r = -0.02$, $p = 0.89$), demonstrating that standard optical inspection fails to capture the defects governing creep failure.
- Post-test interfacial failure percentage strongly correlated with lifetime ($r = -0.67$, $p < 0.001$), indicating that interfacial quality – not bulk void content – determines durability. However, this metric is unavailable before testing and cannot serve as a predictive quality control parameter.
- Three distinct failure mechanisms were identified: Type A (immediate edge delamination, median rupture: 18 s), Type B (progressive edge damage, 207 s), and Type C (bulk-dominated creep, 581 s). Survival analysis confirmed clear separation between types ($p < 0.001$).
- Stress relaxation analysis ($\tau_1 = 5.3$ s, $\tau_2 = 1250$ s) revealed that rapid failures occur before stress redistribution can stabilize the joint, while durable joints survive beyond long-term relaxation timescales.
- SEM imaging revealed sub-optical interfacial imperfections at overlap edges, explaining why optically visible voids do not predict failure. Critical defects are localized at stress concentration sites and invisible to standard inspection methods.

Unlike prior PSA studies focused on peel loading (where the damage is dominated by cavitation phenomena), this shear-loaded geometry shifts the controlling failure mechanism to interfacial quality where stress concentration areas are determinant. These findings are not directly comparable to peel-based studies and highlight the need for load-specific quality control approaches.

5.1. Limitations and future directions

The present study analysed the natural variation of the interfacial quality of a single manufacturing process. The use of different surface treatments, bonding pressures, or adhesive formulations could further enhance the understanding of the relationship between manufacturing processes and the formation of edge defects. However, the use of one PSA formulation and material substrate limits the generalization of the observed effects, although the identified mechanisms could be applied to other acrylic PSA systems.

To fill the gap between post-tests and pre-tests for quality control, future works should explore the characterization of joint's edges through SEM, with multiple cross-sections ($10000\text{--}5000\times$ magnification), to quantify microvoids and interfacial gaps at corners and mid-span areas, and X-ray micro-CT (scan resolution $< 10\text{ }\mu\text{m}$) to map void location and morphology in relation to stress concentration zones. Regarding newly advanced non-destructive techniques, including Lamb wave propagation analysis and Electromechanical Impedance Spectroscopy (EMIS), they could be used as an alternative to evaluate weak adhesion and to detect sub-optical interfacial defects without destructive testing.

Also, X-ray micro-CT would mechanistically validate, during creep testing, the proposed competition between redistribution of viscoelastic stresses and interface crack propagation. Finite Element Analysis (FEA) should be used to assess the shear and peel stress concentrations around the edges and corners and, by comparing them to experimental results correlate them to areas of higher stresses and damage initiation, enabling calculation of critical defect sizes and locations for Type A versus Type C behaviours.

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Author contributions

CRedit: **B. D. Simões:** Investigation, Methodology, Validation, Writing – original draft; **E. A. S. Marques:** Supervision, Validation, Writing – review & editing; **R. J. C. Carbas:** Investigation, Resources, Supervision, Writing – review & editing; **S. Maul:** Conceptualization, Resources, Supervision, Writing – review & editing; **P. Stihler:** Conceptualization, Resources, Supervision, Writing – review & editing; **P. Weißgraeber:** Conceptualization, Supervision, Writing – review & editing; **L. F. M da Silva:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

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